Sensors and Actuators A 271 (2018) 66–75

Contents lists available at ScienceDirect

Sensors and Actuators A: Physical

journal homepage: www.elsevier.com/locate/sna

Sensor fusion, sensitivity analysis and calibration in shooter localization systems

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ARTICLE INFO

Article history:
Received 22 March 2017
Received in revised form 5 October 2017
Accepted 18 December 2017
Available online 2 January 2018

Keywords:
Shooter estimation
Sensitivity analysis
Sensor network
Acoustic array processing

ABSTRACT

This paper analyses the principles and the underlying mathematical model for acoustic based shooter detection systems. These systems use the muzzle blast and acoustic shock waves to compute the shooter's location. Our detection system works on a distributed sensor network where microphone arrays are used as sensors. Detection algorithms run concurrently at each node. After determining the wave directions, the information is passed to the central node. The central node fuses data coming from each sensor with our optimization algorithm. For correct georeferenced fusion, GPS, accelerometer and magnetometer data are used. A mathematical framework of the problem with possible node outputs has been developed in this study. Our framework supports all possible combinations of muzzle blast and shockwave measurements at each sensor node. The simulation for various scenarios has been performed and analyzed. Imperfections in the placements of microphones within the arrays, uncertainty of sensor locations and time uncertainty in time of arrival estimations are investigated as regards to their effect on system performance. In order to analyze the effect of these imperfections on the calculated shooter position, shooter's relative direction and bullet direction, a detailed sensitivity analysis has been done. A system calibration method has been developed in this study to reduce imperfections and enhance the shooter localization performance. In addition, acoustic-based shooter localization hardware has been designed within the study. The study involves not only the simulation but also the results of real tests.

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1. Introduction

In today's border security concept, critical infrastructure protection has a high priority. Especially border posts, personnel buildings and pipelines are the main examples. In protection of these sites and to increase the situational awareness, automated systems are essential. In the case of shooter attack, fast response can increase the survival rate. Therefore, shooter detection systems are developed and used among other systems. Shooter detections systems basically use acoustic and electro-optic sensors. In this paper, we will focus on the use of acoustic sensors.

The purpose of acoustic detection systems is to collect and analyze the data gathered by acoustic sensors to locate the shooter. Acoustic shooter detection systems can be integrated with video surveillance systems or automated weapons to detect and locate the shooter so that the aggressor can be retaliated with quick response to minimize casualty. Thus, it is important to understand the limits of shooter detection systems.

Localization of firearms by detecting and processing the acoustic waves has been in practice since First World War. One of the first systems was an acoustic system called PALS developed in Italy to detect the artillery fires [1].

The range of acoustic sensors are not as high as radar sensors; acoustic wave propagation highly depends on atmospheric conditions. Temperature variations in the path can refract the acoustic waves through propagation. These aspects have to be kept in mind in designing a practical acoustic shooter detection system.

In a typical shooter detection system, there are at least three sensor subsystems which consist of microphone arrays. The embedded hardware inside the sensor subsystems runs the array signal processing algorithms. At the center hub of the systems lies the Data Fusion Center (DFC). DFC collects the data coming from the sensor subsystems apply fusion algorithms and provide the high precision positioning coordinates to the command and control system. The main duty of this central node is to process raw individual sensor measurements to form a more sophisticated and informative fused

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https://doi.org/10.1016/j.sna.2017.12.042
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knowledge. By the help of this data fusion, it is possible to precisely locate the shooter and track its location.

These sensor subsystems also have an inertial measurement system for georeferenced measurements and a GPS receiver for positioning. Synchronization of distributed sensors is critical in sensor fusion of distributed sensor arrays. There are novel synchronization methods in the literature for distributed sensor arrays [2–4]. In the scope of this study, we only used timing output of GPS receivers for synchronization.

Modern systems use multiple microphones and apply array signal processing. ESPRIT algorithm makes frequency domain analysis and uses eigen-decomposition in covariance matrix to find the signal's direction of arrival passively. The con side of these algorithms is the necessity to use too many sensors. Another similar algorithm is MUSIC (Multiple Signal Classification). In this algorithm, narrowband assumption is essential for the success of this algorithm [1].

BBN company has developed a shooter detection system using the TOA (Time of Arrival) of the muzzle blast wave and shock wave in 1996 [5]. They also estimated the caliber of the firearm. For this purpose, they used a microphone array consisting of 4 microphones that are placed in a tetrahedral form. One advantage of using a microphone array is having minimum synchronization error since all the microphones are connected to the same hardware. A basic mathematical model similar to one in BBN for the bullet physics is also given in this paper.

The papers [6,7] use a similar TOA method. But instead of array microphones, they use distributed unsynchronized microphones. At each microphone, muzzle blast and shockwave time of arrival are detected and their time difference is processed at the central node. There is no synchronization problem as only the TOA time differences are processed. But in this method, directional uncertainty increases as the time difference between shock wave and muzzle blast basically gives more information on the range. Because of the directional observability problem, the directional uncertainty becomes quite high.

In [8], a prototype system design resembling [5] has been presented. Low-level muzzle blast and shock wave detection and weapon classification algorithms are discussed. [9] gives a description of the firearm detection system called PILAR that is developed by French 01dB-Metravib company. In the system, wireless 4-microphone arrays are used. TOA methodology is utilized. In the case of 2 sensor systems, their performance metrics are given as $\pm 2^{\circ}$ in azimuth angle uncertainty, $\pm 5^{\circ}$ elevation angle uncertainty and 30% uncertainty in range [10] summarizes a list of performance metrics in a shooter detection system which are azimuth accuracy, elevation accuracy, range accuracy, classification success, etc.

Attitude detection algorithms that can be used in a shooter detection system are given in [11]. Accordingly, accelerometers can find the local gravity direction and determine the locally level plane. In this plane, magnetometers will find the north direction.

In our study, we formed the solution for shooter detection in terms of a sensor network of microphone arrays [12]. Any of the arrays may detect both muzzle blast, and shock waves or just one of them depending on the geometry. Our framework fuses the available information from each sensor. In the literature, there is not enough emphasis on the effects of the modeling uncertainties in the performance of shooter detection system. Our paper shows these effects by our sensitivity analysis. We developed a system calibration model as a result of the sensitivity analysis. This method is used to calibrate uncertainties in the system parameters to enhance system performance. Finally, we implemented our framework with novel MEMS microphones.

2. Shooter localization

2.1. Bullet physics

After a gunfire, two types of waves are formed. First one is the result of the blast while the bullet is traveling the barrel and it propagates spherically at the speed of sound. These sound waves are below 500 Hz and called muzzle blast. The second acoustic wave stems from the air friction of the bullet that is traveling at a multitude speed of sound and this type of wave travels conically around the path of the bullet. This type of wave is called a shock wave. Shock wave has a higher frequency content in 1–4 kHz and characterized as an N shaped signal in the time domain. Fig. 1 shows both muzzle blast and shockwave signals corresponding to a gunshot.

As the muzzle blast sound travels slower than the bullet, the first wave that comes to the point closest to the bullet path is the shock wave. Of course, this can change according to the geometry of sensor, shooter and bullet direction. Shock waves alone are not sufficient to locate the shooter but together with the muzzle blast, they provide crucial information.

2.2. Mathematical model for bullet physics

Acoustics formed by a typical shooting and the sensor geometry is given in Fig. 2. Muzzle blast wave propagates spherically centered at the tip of the muzzle with a sound of speed which depends on atmospheric conditions. Shock waves travel conically around the bullet's path, and the cone angle (θ) depends on the bullet speed as $sin\theta = 1/M$. Here θ denotes the cone angle, M denotes the bullet speed (v) in terms of Mach number (1 Mach = local speed of sound c) where $M = v/c$. Therefore, the more the bullet speed, the less the cone angle. A supersonic bullet speed can range from 1–5 M.

In Fig. 2, x_0 shows the point where the bullet leaves the muzzle, at time t_0. Muzzle blast wave propagates circularly from that point. Muzzle blast sound arrives at the sensor subsystem s_1 at time t_{n1} whereas the shock wave arrives at time t_{s1}. The time instants mentioned here are ToA's at the reference microphone at sensor subsystem. Shock wave that first comes to the sensor at s_1 is formed at x_1; bullet comes to this point at a higher speed than the sound speed, after that point it travels at 1 Mach.

By using TOA algorithms as mentioned above, the muzzle blast and shockwave incoming directions u_1, u_2 can be calculated. In our setting:

$$\begin{aligned} x_0 &= x_2 + c\ u_2(t_{s1} - t_0) \\ x_0 &= s_1 + c\ u_1(t_{n1} - t_0) \end{aligned} \tag{1}$$

Then,

$$s_1 + c\ u_1(t_{n1} - t_0) = x_2 + c\ u_2(t_{s1} - t_0) \tag{2}$$

As s_1x_2 is perpendicular to u_2, t_0 can be found as below:

$$\begin{aligned} &(x_2 - s_1)\cdot u_2 = 0 \\ &(-c\ u_2t_{s1} + c\ u_2t_0 + c\ u_1t_{n1} - c\,u_1t_0)\cdot u_2 = 0 \\ &t_0 = \frac{t_{s1} - t_{n1}u_1\cdot u_2}{1 - u_1\cdot u_2} \end{aligned} \tag{3}$$

Shooter location x_0 can be calculated from Eq. (1) by using t_0 calculated from Eq. (3). A similar result can be found in [8]. When there is only one sensor subsystem, bullet path cannot be computed. The bullet path can be computed when the path lies between two sensor subsystems. In single sensor case, shooter position can be determined because both shock wave and muzzle blast directions and timings are available.

In fact, this setting assumes that the cone angle is constant and the bullet propagates at a constant speed which is not exactly cor-

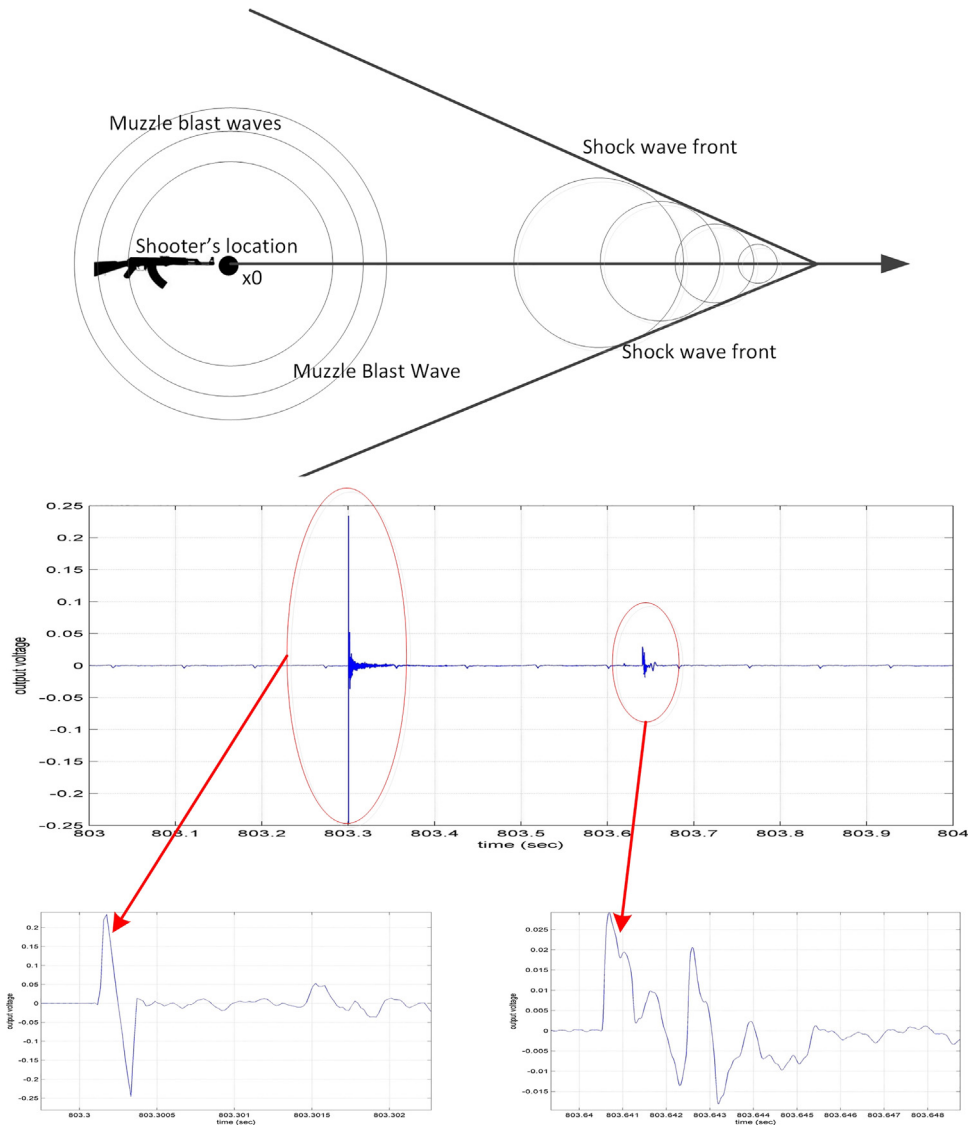

Fig. 1. Shock wave and muzzle blast wave [9].

rect in reality. In reality, the cone angle increases as the bullet speed decelerates due to air friction. Thus, we need a comprehensive treatment for multi-sensors. In addition, the simulation and field test results show that performance of the system is adequate with those assumptions.

Now let's calculate x_1, the starting point of the shock waves coming to point s_1 and its time of arrival t_{s1}. u_0 shows the unit vector for the bullet direction of travel, u_1 is the unit vector for the muzzle blast direction of arrival and u_2 is the unit vector for the shock wave direction of arrival. Up to this point, the measured parameters are x_0, u_0, t_0, s_1. Then the point x_3 which is the closest point on the bullet path to the sensor subsystem can be written as:

$$x_3 = [(s_1 - x_0) \cdot u_0]\, u_0 + x_0 \tag{4}$$

Using the cone angle θ which can be estimated by the bullet speed,x_1 can be found. Here let α be the scalar length of the vector $(x_1 - x_0)$. Then,

$$\begin{aligned} x_1 &= x_0 + \alpha u_0 \\ [x_3 - (x_0 + \alpha u_0)] \cdot u_0 &= |s_1 - x_3| \tan\theta \\ \alpha &= x_3 \cdot u_0 - x_0 \cdot u_0 - |s_1 - x_3| \tan\theta \end{aligned} \tag{5}$$

In this case, the time t_{s1} becomes,

$$t_{s1} = |x_1 - x_0| / (M\ c) + |s_1 - x_1| \tag{6}$$

Muzzle blast time t_{n1} can be stated as:

$$t_{n1} = |s_1 - x_0| / c + t_0 \tag{7}$$

Using these variables and deductions, the next step is to calculate muzzle blast direction (u_1), muzzle blast time (t_{n1}), shock wave direction (u_2) and shock wave time (t_{s1}) in terms of unknown variables x_0, u_0, t_0 and known sensor positions s_i. In summary, the dependency of the variables in the problem can be written in a nonlinear equation form as follows:

$$\xi \overset{\Delta}{=} (t_{n1}, t_{s1}, u_{n1}, u_{s1}, t_{n2}, t_{s2}, u_{n2}, u_{s2}, \cdots)^T = f(t_0, x_0, u_0) \tag{8}$$

$t_{si}, t_{ni}, u_{ni}, u_{si}$ are calculated using the arrays and we can define the following optimization:

$$\min_{t_0, x_o, u_o} \left[(\xi - f)^T \left(\xi - f \right) \right] \tag{9}$$

Eq. (9) can be solved by nonlinear least squares optimization techniques to find $x_0,\ u_0,\ t_0$ Notice that this optimization can be set up

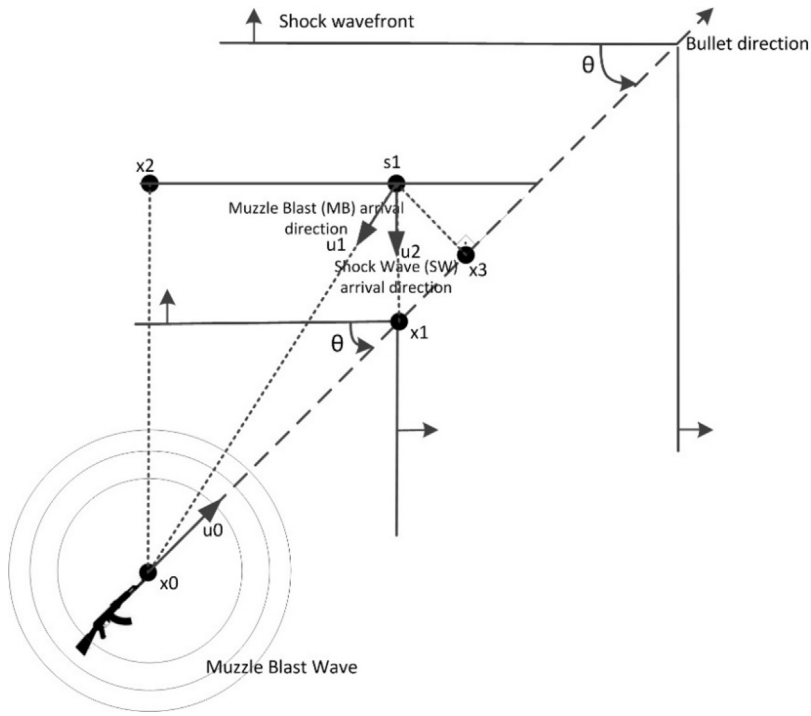

Fig. 2. Modelling of bullet physics and sensor geometry in a shooting.

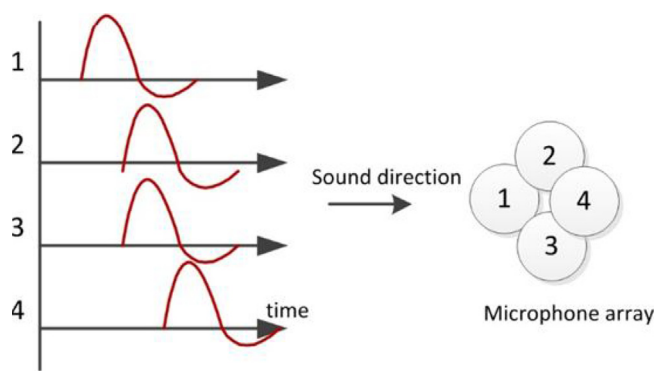

Fig. 3. Direction finding through TOA of the sound signal.

for any number of sensors. This optimization is quite fast, it takes only 20 iterations to achieve reasonable error levels.

2.3. TOA algorithm

Using the time of arrival of the sound signal at the microphone array it is possible to find the direction of arrival. In the case of gunfire, using this methodology both muzzle blast and shock wave direction arrivals can be found. As shown in Fig. 3, the microphone closest to the gunfire source senses the sound first. If one can write the relevant equations, it can be solved using a least squares algorithm.

Let's assume that we have four microphones and their positions are given by z_i, then TDOA (Time Difference of Arrival) of sound waves (Δt) can be found using the local clock. The direction of arrival u can be found as

$$\begin{bmatrix} (z_2 - z_1)^T \\ (z_3 - z_1)^T \\ (z_4 - z_1)^T \end{bmatrix} u = Au = -\Delta t \cdot M \tag{10}$$

Then the u can be found as

$$u = \begin{cases} -MA^{-1}\Delta t & \text{with 4 microphones} \\ -M(A^TA)^{-1}A^T\Delta t & \text{with more than 4 microphones} \end{cases} \tag{11}$$

In general, if we also require the unity norm for u as in Eqs. (4)–(9), we can use the quadratic-eigenvalue QEP method as in [13] ($\min J(x) = |A * x - b|^2$ s.t.$|x| = 1$).

2.4. Sensor fusion

Each sensor subsystem may detect both muzzle blast and shock wave or it may detect either one of them. Whichever the case in each sensor is, our algorithm solves whatever the inputs are and finds the solution through optimizing Eq. (9). Although we used the local clocks in the sensors to calculate the DOA of the waves, we use the GPS time output for the synchronization of multi-sensors (t_{si}, t_{ni}) in the fusion process. The solution of the fusion algorithm is given in Table 1.

2.5. Signal processing algorithm

The purpose of signal detection algorithm is to extract gunshot waves (SW and MB) from noisy acoustic signals and provide time-of-arrival (TOA) of this waves to direction finding algorithm. Although gunshot acoustic signals (SW and MB) are both transient and impulsive, they have different characteristics to distinguish from each other. Typical shockwave has sharp rising and falling edges and N wave type and lasts 200–300 μs [8]. The muzzle blast has slower rise and falling edges as compared with shockwaves. The muzzle blast lasts 3–5 ms and its first half period is shorter than the second one. The recorded bullet shock waves and muzzle blast are shown in Fig. 1.

Signal detection algorithms are used to extract gunshot shock wave and muzzle blast from recorded real gunshot signals. First, the start and end of any acoustic events at an incoming signal are marked roughly at activity detection block according to adaptive thresholding. Each marked activity region is divided into segments

Table 1
Fusion Performance.

Detected Wave Type	1 × Sensor subsystem	2 × Sensor Subsystems	3 × Sensor Subsystems
Only Shock Wave	Location or direction of the source cannot be computed.	Direction can be computed via the bullet trajectory. No location.	Direction can be computed via the bullet trajectory with higher accuracy. No location
Only Muzzle Blast	Direction can be computed. No location.	Location can be computed. The accuracy depends on the geometry.	Location can be computed with higher accuracy. The accuracy depends on the geometry.
Both Shock Wave & Muzzle Blast	Location and direction can be computed.	Location and direction can be computed with higher accuracy.	Location and direction can be computed with higher accuracy.

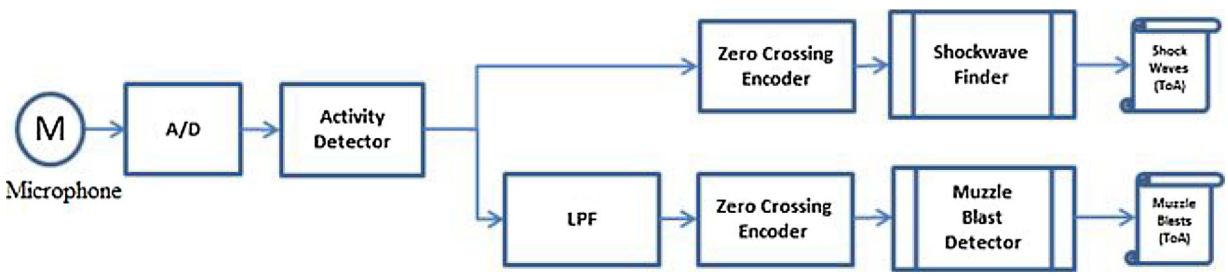

Fig. 4. Signal Detection Algorithm Blocks.

at zero crossing encoder which is commonly used voice detection [14]. Start time, rise time, peak amplitude, peak time and stop time values are computed and stored for each segment between zero crossings for later processing. Zero encoded segments are analyzed and compared with shockwave or muzzle blast characteristics. If there is a proper match then it is labeled as SW or MB. The block diagram of the detection algorithm is shown in Fig. 4. It has been observed that the signal detection algorithm could detect 96% of the recorded real gunshot signals.

3. Sensitivity analysis

Sensitivity analysis has been done using the mathematical modeling given in Section 2. As the mathematical equations are highly nonlinear we preferred to make Monte Carlo (MC) simulations to find out sensitivity characteristics to several input variables. These are:

i Bullet speed uncertainty,
ii Sound speed uncertainty,
iii Distance uncertainty between microphones in the array,
iv TOA data uncertainty due to sampling of the microphone data,
v All the errors combined together,
vi Usage of 1 sensor vs. 2 sensors.

In each sensitivity analysis, 500 MC simulation runs have been performed. The position of the shooter is (500, 500, 0) in (x, y, z) coordinates with the unit in meters. The shooter aims at (0, 0, 0), Sensor 1 is located at (100, 0, 0) and sensor 2 is located at (0, 200, 0). Nominal bullet speed is 3.73 M and default speed of sound is taken as 343 m/s.

3.1. Bullet speed uncertainty

For bullet speed uncertainty, randomly selected bullet speed is used according to the following model:

$$M = 3.73Mach + 0.373Mach \cdot N(0, 1) \tag{12}$$

where $N(0, 1)$ denotes zero-mean unit variance white Gaussian noise. In the estimation calculations, only the nominal values are used. The corresponding computed shooter locations are given in Fig. 5(a). 500 MC simulations have been done to get the RMS error in horizontal position. Errors in the bullet speed mainly affect the range of the shooter. Interestingly when 2 sensors are used the error increases. This may be due to the fitting of the data to two different sensor locations and the geometry in optimization.

3.2. Sound speed uncertainty

Sound speed depends on atmospheric conditions in general. Temperature, pressure are the main contributors for the value of the sound speed.

$$1Mach = 343 + 10 \cdot N(0, 1)m/s \tag{13}$$

The corresponding computed shooter location estimations are given in Fig. 5(b). Errors in the sound speed mainly affects the range of the shooter. When 2 sensors are used the error decreases. If we also measure the temperature and make a better estimate for the sound speed, potentially it will have an important effect on the performance

3.3. 1 cm microphone location error in the array

Four microphones in the sensor array are placed to form a tetrahedral. As the measurements are relative to the reference microphone in the sensitivity analysis a 1 cm uncertainty is added to the three other microphones. The sensitivity analysis results can be seen in Fig. 5 (c). The errors in shooter location are in both azimuth and range at the same time. Again, using two sensors improves the results considerably.

3.4. 2 cm microphone location error in the array

As the measurements are relative to the reference microphone in the sensitivity analysis, a 2 cm uncertainty is added to the three other microphones. The sensitivity analysis results can be seen in Fig. 5(d). The errors in shooter location increased in both azimuth and range at the same time. Again, using two sensors improves the range results considerably. But not as much for the azimuth uncertainty.

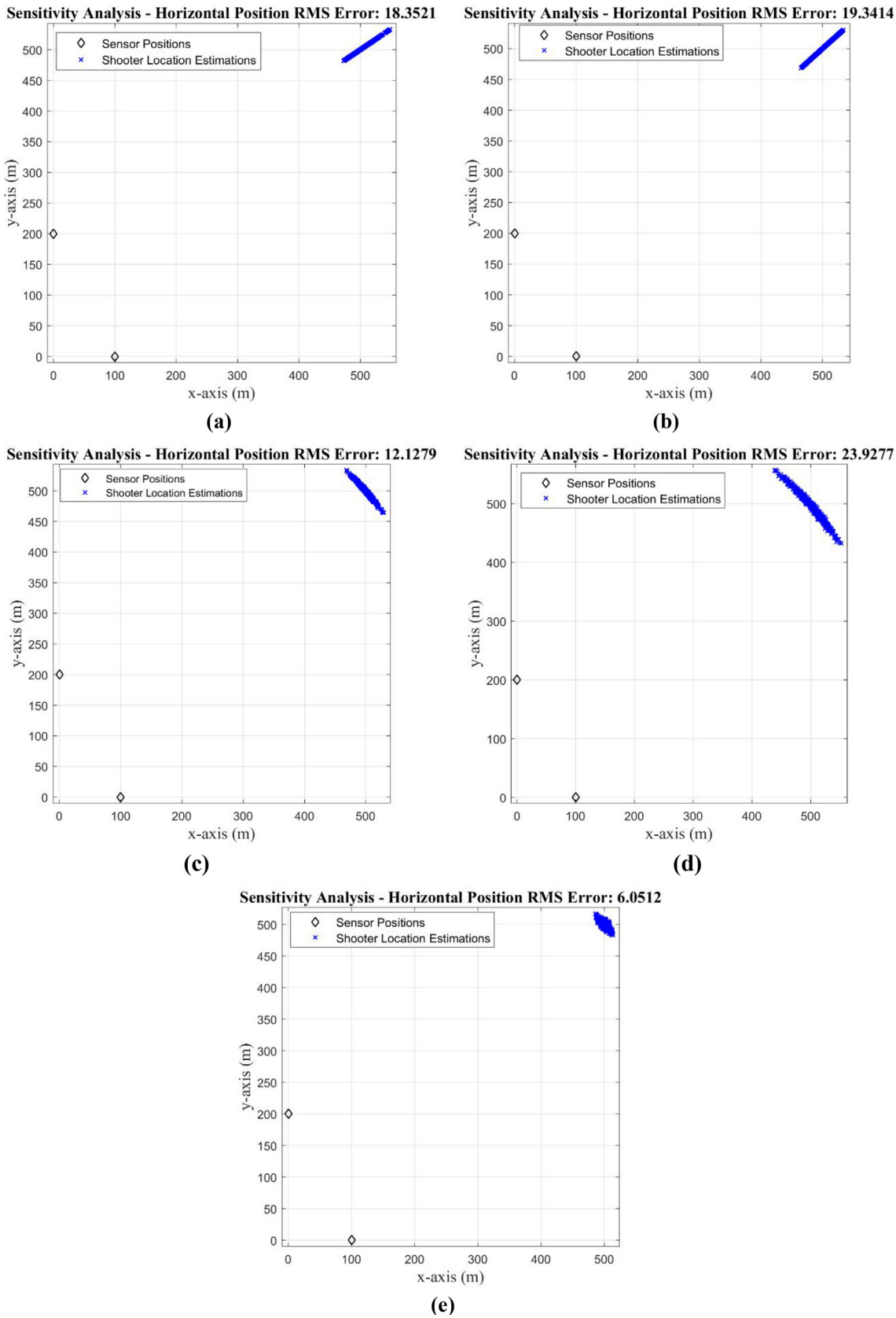

Fig. 5. Sensitivity analysis to various factors for 2 sensors.

Table 2
Horizontal position error (m) after 50 MC simulations.

System Parameters	1-sensor system	2-sensor system
Bullet speed (10% error)	12.34	18.35
Sound speed (10 m/s error)	31.82	19.34
1 cm 1σ microphone positioning error	28.05	12.13
2 cm 1σ microphone positioning error	51.23	23.93
TOA error uniformly distributed over U(0,19.61 μs)	13.76	6.05
With all error sources (Bullet speed, Sound speed, Mic. Pos., TOA)	76.10	45.92

3.5. TOA uncertainty

Microphone data is sampled at 51.2 kHz. Therefore, the sampling period is 19.61 μs. That means in the detection process a uniform error 0–19.61 μs is possible. So, this time we added such an uncertainty to timings in calculating the muzzle blast and shock wave directions. The results can be seen in Fig. 5(e).

3.6. All the error sources combined

When we combine all the errors, which are 10% bullet speed error, 10 m/ssound speed error, 2 cm microphone placement error, U [0, 19.61 μs] error in TOA timings, shooter location distribution takes the shape as in Fig. 6.

The errors in shooter location estimations for one sensor and two sensors are given in Table 2.

4. System calibration

Shooter localization depends on system parameters such as sound speed (c), microphone positions (ΔMic_{ARR}), and bullet speed (M). These parameters are generally determined by measurement devices or estimations. However, these parameters might be erroneous. For instance, the sound of speed varies with the ambient temperature; microphone positions vary due to thermal expansion and contraction, and bullet speed is a usually unknown parameter and estimated within a reasonable range. Erroneous values of system parameters affect the system performance. Thus, a system calibration method is proposed to improve estimation accuracy [15].

4.1. System calibration method

The calibration method is applied to measurements regarding a controlled shot in which locations of shooter and target are known make a better estimation of sound speed (c), microphone positions (ΔMic_{ARR}), and bullet speed (M). The calibration method is as follows:

1 Known parameters are shot location x_0, target location T_0, and sensor location s_0,
2 Initial values for ΔMic_{ARR},$M$and c are set, where $\Delta Mic_{ARR} = \begin{bmatrix} z_2 - z_1 \\ z_2 - z_1 \\ \vdots \\ z_N - z_1 \end{bmatrix}, \begin{pmatrix} N : Microphone\ Number \\ 1\ : Reference\ Microphone \\ z_n : microphone\ position \end{pmatrix}$,
3 Unit direction of bullet u_0 is calculated directly by $u_0 = \frac{T_0 - x_0}{|T_0 - x_0|}$,
4 Unit direction of muzzle blast u_n is calculated by $u_n = \frac{x_0 - s}{|x_0 - s|}$,
5 TDoA of muzzle blast is calculated by $\Delta\hat{t}_n = \Delta Mic_{ARR}(\frac{-u_n}{c})$, and the cost function is $\tilde{J}_{\Delta t_n} = \left|\Delta t_n - \Delta\hat{t}_n\right|^2$
6 Unit direction of shockwave $\hat{u}_s$ is obtained by $u_s = \frac{x_s - s}{|x_s - s|}$ where $\hat{x}_s$ is obtained by (5),
7 TDoA of shockwave is calculated by $\Delta\hat{t}_s = \Delta Mic_{ARR}(\frac{-u_s}{c})$ and the cost function is $\tilde{J}_{\Delta t_s} = \left|\Delta t_s - \Delta\hat{t}_s\right|^2$
8 Shot time $\hat{t}_0$ is calculated by (3),
9 Shockwave time $\hat{t}_s$ is calculated by (6) and the corresponding cost function is $\tilde{J}_{t_s} = \left|t_s - \hat{t}_s\right|^2$,
10 Muzzle blast time $\hat{t}_n$ is calculated by (7) and the cost function is $\tilde{J}_{t_n} = \left|t_n - \hat{t}_n\right|^2$,
11 $\hat{x}_0$ is calculated by (1) and the cost function is $\tilde{J}_{x_0} = \left|x_0 - \hat{x}_0\right|^2$,
12 Single cost function is obtained with weighted sum of cost functions in preceding steps

$$\begin{aligned}
&\min_{c, M, \Delta Mic_{ARR}} J(c, M, \Delta Mic_{ARR}) = \tilde{J}_{\Delta t_n}(M, \Delta Mic_{ARR}) + \tilde{J}_{\Delta t_s}(c, M, \Delta Mic_{ARR}) + \tilde{J}_{t_n}(M) + \tilde{J}_{t_s}(c, M) + \tilde{J}_{x_0}(c, M, \Delta Mic_{ARR}) \\
&subject\ \ to: \\
&1 < \hat{M} < M_{MAX} \quad , \quad M_{MAX} = 5 \\
&\Delta Mic_{ARR} - Err_{\Delta Mic_{ARR}} < \Delta\hat{M}ic_{ARR} < \Delta Mic_{ARR} + Err_{\Delta Mic_{ARR}} \ , \ Err_{\Delta Mic_{ARR}} = 0.03 \times \begin{pmatrix} 0 & \cdots & 0 \\ 1 & \cdots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \cdots & 1 \end{pmatrix}_{NxN} \\
&c \times (1 - E_c) < \hat{c} < c(1 + E_c) \quad , \quad E_c = 10\%.
\end{aligned} \tag{14}$$

Nonlinear constrained optimization problem stated in (14) is solved with interior-point method [16] to estimate system parameters of shooter localization system. Constraints are determined by observations and measurements. Bullet speed ($\hat{M}$) is assumed to be between 1 Mach and 5 Mach; microphone array position ($\Delta\hat{M}ic_{ARR}$) is assumed to be between error at each coordinate, and sound speed ($\hat{c}$) is assumed to be in the range of 10% percentage error.

4.2. System calibration test results

System calibration method is performed on simulated data of a single array of eight microphones which is placed at a fixed location. The reference microphone on the array is set to the origin of the coordinate system. The coordinates of target and shooter locations are determined with respect to the origin. Actual system parameters of the simulation are listed in Table 3.

In the simulation, position measurement error is assumed as a normal distribution of $N(0, 10\,m)$ and applied to the positions of the sensor, target, and shooter. Microphone time measurement error is

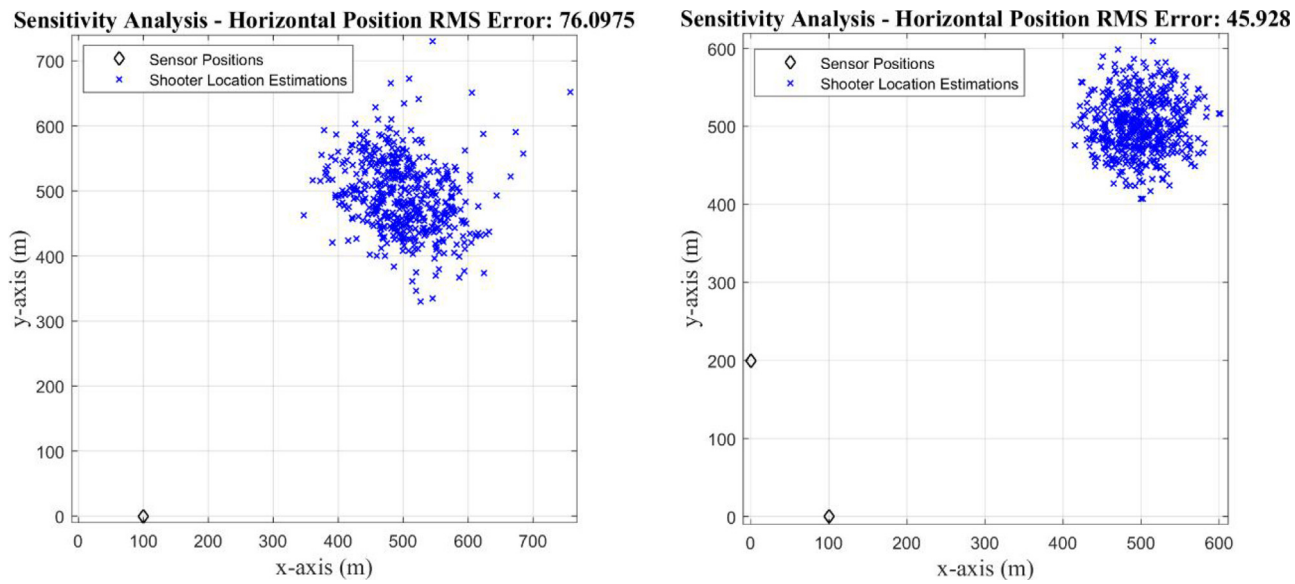

Fig. 6. Sensitivity analysis to all the error sources combined for 1 sensor (left) and 2 sensors (right).

Table 3
Shooter Localization System Parameters for Sample Shooting Test.

Sound Speed	Bullet Speed	Microphone Array Positions
$c = 340\ m/s$	$M = 2.1\,Mach$	$\begin{array}{l} Mic_1: \\ Mic_2: \\ Mic_3: \\ Mic_4: \\ Mic_5: \\ Mic_6: \\ Mic_7: \\ Mic_8: \end{array} \begin{bmatrix} 0 & 0 & 0 \\ 0.4 & 0 & 0 \\ 0.2 & 0.2 & 0 \\ 0.2 & -0.2 & 0 \\ 0.1333 & 0.0667 & 0.2828 \\ 0.2667 & -0.0667 & 0.2828 \\ 0.1333 & -0.0667 & 0.2828 \\ 0.2 & 0 & 0.4 \end{bmatrix} \begin{array}{l} \mapsto \textit{reference microphone} \\ \\ \\ \\ \\ \\ \\ (units: m) \end{array}$

Table 4
System Parameter Estimation for 250 MC simulation.

System Parameters	Estimation Error Standard Deviation
Sound Speed	9.1096 m/s
Bullet Speed	0.0688 Mach
Microphone	6.5898 mm

Table 5
Estimation Performance for 250 MC simulations.

Shooter Location Estimation	Estimation Error Standard Deviation
Uncalibrated System	104.5055 m
Calibrated System	4.5388 m

assumed as uniform distribution between $\left[-T_s/2, T_s/2\right]$ where T_s is sampling period ($F_s = 44100 Hz T_s = 1/F_s \Rightarrow T_s = 22.6\ \ 10^{-6}$ sec) and applied to TOA and TDOA of shockwave and muzzle blast. Microphone position error is assumed as $N(0, 0.03\,m)$. Speed of sound percentage error is assumed as a normal distribution of $N(0, 0.1)$. Erroneous bullet speed measurement is assumed as uniform distribution between [1, 4].

The system calibration method is performed on 250 Monte Carlo simulations to calibrate system parameters. Results of system calibration simulation for the system parameters and shooter location estimations corresponding to uncalibrated and calibrated parameters are listed in Tables 4 and 5, respectively.

5. Hardware design

Shooter localization system hardware is designed to

- Process maximum number of eight microphone signal simultaneously,
- Measure position and orientation of the system in order to estimate geographical coordinates of shooter location,
- Be used as a sensor node in wireless network,
- Store microphone signal data and system configuration parameters,
- Be powered by either external source or internal battery.

Shooter localization system hardware design is composed of commercial off the shelf hardware and genuine hardware design.

In the final design of shooter localization hardware, CPU (Analog Devices - BF707) used in the shooter localization has the capability of processing microphone signals, orientation, and geographical location simultaneously in order to run shooter localization algorithms.

Up to eight microphone signals can be filtered and amplified in signal conditioning unit. The output of signal conditioning unit is converted to a digital signal in ADC (Analog Devices – AD8568) unit which has 16-bit resolution and a sampling frequency of 44.1 KHz. The bandwidth of digital signal for eight microphones is $44100\ Hz \times 8 \times 16\,bit = 5,644,800\ bps$.

There are several auxiliary units providing system physical information required for estimation of shooter location such as inertial measurement unit and GPS. Position and orientation of the sensor node are measured in the commercial of the shelf iner-

Table 6
Shooter location estimation results for real shots.

Actual Shooter Location (m)			Shooter Location Estimation (m)			Range Error		Azimuth Error	Elevation Error
x	y	z	x	y	z	(m)	%	(°)	(°)
49	−5	0	46.0527	−5.3323	−2.6859	6.13	12.52	−0.77	−3.3
123	−5	0	119.5710	−1.7063	4.7696	10.16	8.13	1.51	2.3
123	−5	0	122.8449	−9.5000	38.8991	38.96	31.16	−2.13	17.5
123	−5	0	114.5219	−4.6836	18.7084	22.09	17.67	−0.05	9.2
123	−5	0	122.8800	2.4310	−1.8778	12.74	10.19	3.42	−0.8

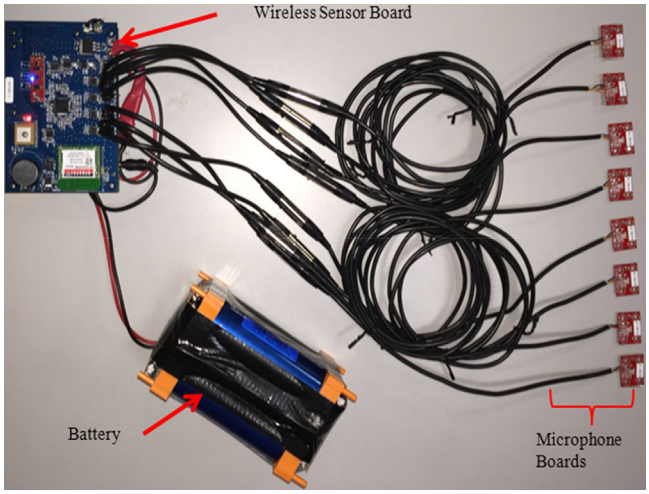

Fig. 7. Shooter localization hardware.

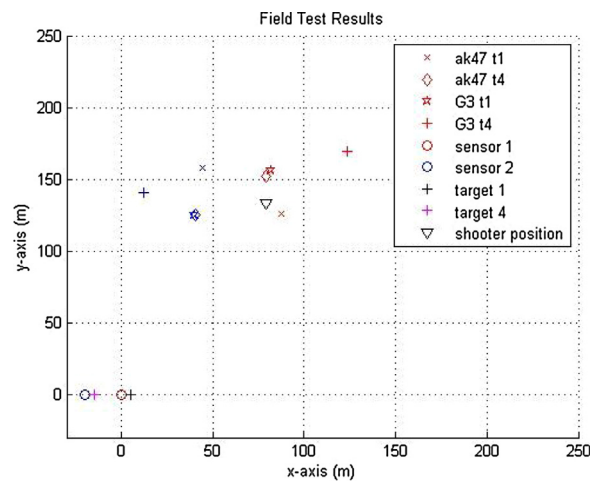

Fig. 8. A real shot and shooter location estimation. (For interpretation of the reference to colour in text, the reader is referred to the web version of this article.)

tial measurement unit (SPARKFUN-Sensor Stick) and geographical coordinates are measured by GPS (GTOP Ladybird PA6C).

Additionally, there are supplementary system features in product level. There is an RF unit (SYNAPSE RF Module) in order to establish communication between shooter localization and remote command and control center. There is also data storage unit on the system for offline data process and system diagnostics. The hardware used in shooter localization is shown in Fig. 7.

6. Field tests

Field tests were conducted with a prototype system in a controlled environment with the assistance of the professional snipers. Three acoustic sensor nodes were placed such that it will cover and protect the critical infrastructure. One central unit for surveillance was configured in the command and control station. All sensor nodes are connected wirelessly with command and control unit via synapse sensor network. All system settings can be made remotely via the central node. Furthermore, all microphones in the sensor array were connected to the same time source for overcoming time synchronization problems.

After applying the localization algorithms and fusion models explained in this paper to the gathered real data, we have localized the sniper locations with success in ±2 error in azimuth, %10 error in range. This performance measurement is comparable with the is comparable with PILAR performances [9].

In Fig. 8, you can see that real shotgun localization estimations shot from the same position to the target at 150 m using AK-47 and G3 rifles. In Fig. 8, the sensor arrays placed in 20 m spaced are marked with a circle (◯), the real shot source is marked with a triangle (Δ) and targets which shot through are marked with a plus (+). The shotgun localization estimations calculated sensor 1 and sensor 2 are marked with a red or blue cross (x, x) or red or blue diamond (◊, ◊), respectively. In general, even though it seems a successfully fused localization estimation in terms of direction and range, the biased error in estimated locations and sensor locations implies that the system needs some calibration methods and sensitivity analysis. In the upcoming tests, theoretical and mathematical results of the sensitivity analysis and calibration methods will be tested and shown on the real data collected via field tests.

Table 6 lists shooter location estimation results corresponding to several real shots located at 120m and 50 m away from the sensor deployed in the field.

7. Discussion and conclusion

In this paper, we explored the factors that affect the performance of an acoustic shooter detection system. As the sensitivity analysis in Table 2 points out, the bullet speed depends on the rifle and the cartridge caliber. If it can be classified then the error due to bullet speed will decrease and has a good impact on the estimation accuracy. Better estimation of local sound speed will also improve the results, so a thermometer at the sensor subsystems will help to decrease the estimation error.

The placement accuracy of the microphones in the array seems very important. A careful measurement with less than 1 cm accuracy will be very helpful. In the field, temperature variations may vary and the inter-distance between the microphones may change, so this sensitivity analysis shows that a self-calibration or a system identification process would be beneficial for the performance of the system. 51.2 kHz sampling rate does not add too much error in the estimates. Thus, any rate faster or equal to 51.2 KHz seems fine. When we add all the error sources the results seem to match real systems which are 10% error in the range 2° accuracy in the azimuth.

In this study, we developed a realistic mathematical model for an acoustic based shooter detection system, and we analyzed the effects of several error sources on the position estimation of the shooter. This analysis is important in tradeoff decision process in

the design of such a system. In our framework, it is also possible to deal with all combinations of SW and MB measurements in the sensor network which increases the performance of the fusion process.

Acknowledgements

We would like to thank Turkish Military Academy Defense Sciences Institute for providing us military shotguns test field and Ministry of Science, Industry, and Technology of Turkey for the financial support of 0291.STZ.2013-2 coded SANTEZ project.

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Biographies

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